Financial Constraints and Trade Credit as a Strategic Tool: Evidence from Small-Scale Reservation Reforms in India

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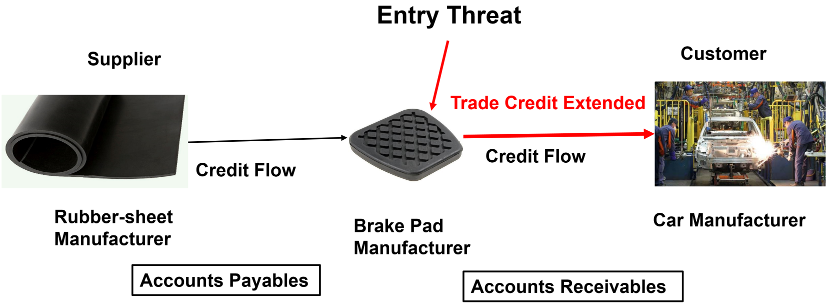
Georgia Tech, Atlanta

December 18, 2017
EMF 2017, Mumbai

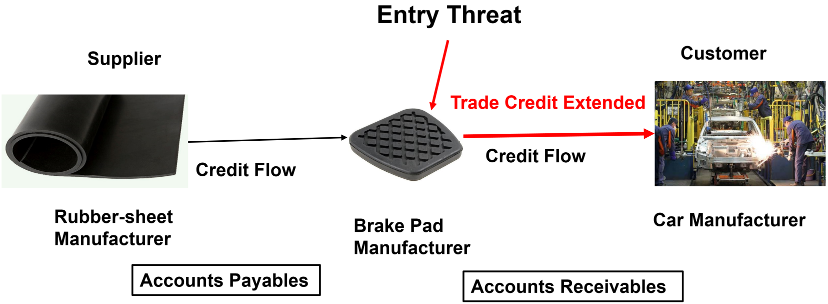
Example


Entry Threat
Supplier
Customer
Trade Credit Extended
Credit Flow
Credit Flow
Rubber-sheet Manufacturer
Brake Pad Manufacturer
Car Manufacturer
Accounts Payables
Accounts Receivables

Example

Research Question

Do firms extend more credit to their customers when they face entry threat from competitors or cut down price or both ?



Motivation

How Incumbent Firms Respond ?

Theoretical literature offering rationale for pre-emptive actions (Dixit, 1979)

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  - Limit pricing be cannot used by firms with lower profitability
  - Mature firms (with lower cost of capital) may not prefer to cut prices

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Effectiveness of Pre-emptive Actions

- Costly for potential entrants to replicate
- Customers should value incumbent's action

Motivation

Why Trade Credit ?

- According to Financial Times, in 2007, 90% of world merchandise trade was financed by trade credit to the value of $14 trillion (Williams 2008)
- 80% of US manufacturing firms offer their product on trade credit (Tirole, 2006)
- Accounts receivable (trade credit granted) support 39.2 percent of total sales (Antràs and Foley, 2015)
- In underdeveloped formal credit markets trade credit extended by firms serves as substitute of bank finance (Demirguc-Kunt and Maksimovic, 2002; Fisman and Love, 2003)

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Trade Credit as Strategic Tool

- How Trade Credit Supply affects Industry Dynamics ?
- **French** Trucking Industry: After **exogenous restriction** on the **trade credit supplied** (Barrot, 2016)
  - Higher entry rate /Lower bankruptcy rate for financially constrained firms.

Motivation

Threat of Entry Vs Actual Entry

Interesting : a) Ex-ante response, b) influence entry decision and c) no change in current cash flows/profitability

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Empirical Challenge

Identifying the Threat of Entry separately from Actual Entry

- Firms' trade credit policies and product market strategies are jointly determined in equilibrium
- Shock has to ideally affect the threat of entry, as opposed to actual entry by a competing firm

Reservation of Products

भारत का राजपत्र
The Gazette of India

सं० 412] नई दिल्ली, वृहस्पतिवार, जुलाई 25, 1991/श्रावण 3, 1913
No. 412] NEW DELHI, THURSDAY, JULY 25, 1991/SRAWANA 3, 1913

LIST OF ITEMS RESERVED FOR EXCLUSIVE MANUFACTURE IN SMALL SCALE SECTOR

उत्पाद का नाम	क्रम संख्या (राजपत्र अधिसूचना के अनुसार) Sl. No. (As per gazette notification)	उत्पादन—कोड PRODUCT CODE	NAME OF THE PRODUCT
1	2	3	4
***खाद्य और सहबद्ध उद्योग		20-21	***Food and Allied Industries
आइसक्रीम	1	201801	Ice Cream
	2	Deleted	
आचार और चटनी	3	202501	Pickles & chutneys
सिरका	4	202901	Vinegar
चावल कटाई	5	204200	Rice Milling
दाल तैयार करना (दालों की दलाई)	6	204300	Dal Milling
ब्रेड/डबल रोटी/पाव रोटी	7	205101	Bread
बिस्कुट	8	20530101	Biscuits
पेस्ट्री	8A	20530102	Pastry
मिष्ठान (चाकलेट को छोड़कर)	9	20920101	Confectionery (Excluding Chocolates)*
रेपसीड तेल (निकाले हुए तेल को छोड़कर) सरसों का तेल (सोलवेन्ट निकले तेल को छोड़कर) तिल का तेल (सोलवेन्ट निकले तेल को छोड़कर) — राज्य कृषि सहकारी समितियों और उत्पादक सहकारी समितियों के मामलों को छोड़कर	10	21100101	Rapeseed Oil except solvent extracted — Other than case of state Agri Co-operatives and Growe Co-operatives
	11	21100102	Mustard Oil except solvent extracted — Other than case of state Agri Co-operatives and Growe Co-operatives
	12	21100103	Seasame/oil/except solvent extracted
मूंगफली का तेल (सोलवेन्ट निकले तेल को छोड़कर)	13	21100104	Ground nut oil except solvent extracted
काजू के मीठे उत्पाद	14	214302	Sweetened Cashewnut products
मुर्गीदाना	15	21620101	Poultry feed except in pellet form
पिसे हुए तथा तैयार शुद्ध मसाले	16	21920101	Ground and processed spices**
टैपियोका /साबूदाना	16A	219502	Tapioca sago
टैपियोका/ आटा	16B	219504	Tapioca flour
सिन्थेटिक सिरप	16C	224302	Synthetic syrups
*होजरी सहित कपड़े की बनी चीजें		26	*Textile products including hosiery
बुने हुए सूती कपड़े	17	260101	Cotton cloth knitted
सूती बुने बनियानें	18	260102	Cotton vests knitted
बने हुए सूती मोजे	19	260103	Cotton socks knitted

More than **800 items** which constitute **1024 products** reserved

Examples of Reserved Products

Ice cream

Bus Seats

Hydraulic Jacks

Flooring Tiles

Brake Pad

Barrels

Why Reservation ?

- **Objective:** Absorb the abundant labor supply/ save small firms
- **Entry Barrier:** Exclusive production by small firms (employment/investment ceiling)
- **Existing Firms:** Constraint on creation of fresh capacity in the large/medium sector
  - Export obligation of a minimum of 75% of its production from fresh capacity (Mohan, 2002)
- **Randomness** in selection of products
  - No official documentation that describing how reservation policy was formulated (Garcia-Santana and Pijoan-Mas (2014).
- **Expert Committee:** Abid Hussain Committee (1997) "the choice of products for reservation was necessarily arbitrary"
  - Reserved: Cotton and woolen socks, scarfs, cloths and vests
  - Not Reserved: Linen, jute or hemp textile products
- **Importance:** In 1991, 1024 reserved products constitute 12% of Indian manufacturing output
  - Heterogeneity across industry sectors (Tiwari and Wilde (2014))
  - 80% of output in hosiery and garments, 57% in certain wood products

Dereservation of Products

जिस्ट्री सं॰ डी॰ एल॰–33004/97 REGD. NO. D. L.-33004/97

भारत का राजपत्र

The Gazette of India

असाधारण

EXTRAORDINARY

भाग II—खण्ड 3—उप-खण्ड (ii)

PART II—Section 3—Sub-section (ii)

प्राधिकार से प्रकाशित

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No. 227] NEW DELHI, THURSDAY, APRIL 3, 1997/CHAITRA 13, 1919

MINISTRY OF INDUSTRY

(Department of Industrial Policy and Promotion)

NOTIFICATION

New Delhi, the 3rd April, 1997

S. O. 298 (E).—In exercise of the powers conferred by sub-section (2A) of section 29B of the Industries (Development and Regulation) Act, 1951 (65 of 1951) and after considering the recommendations made by the Advisory Committee constituted under sub-section (2B) of the said section, the Central Government hereby direct that the following amendments shall be made in the notification of the Government of India in the Ministry of Industry, Department of Industrial Development No. S.O.477(E), dated the 25th July, 1991, namely :—

In the said notification, in Schedule-III under the heading—LIST OF ITEMS RESERVED FOR EXCLUSIVE MANUFACTURE IN SMALL SCLAE SECTOR, the following items and the entries relating thereto shall be omitted, namely :—

Sl. No. (As per gazette Notification)	Product Code	Name of the product	
1	2	3	
"1	201801	Ice Cream	
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15	21620101	Poultry feed except in pellet form	
16C	224302	Synthetic syrups	
54	280906	Corrugated paper & boards	
571	363708	Hair driers—all types	
629	374717	Hub caps—auto	
635	374737	Ornamental fittings—auto	
644	37478101	Spot lamps assembly—auto	Excluding combination lamp assembly
645	37478201	Stop lamps assembly—auto	Excluding combination lamp assembly
646	3747870	Tail lamp assembly—auto	
647	3747804	Ash trays—Car fittings"	

Why Dereservation?

- **Advisory Committee on Reservation:** Meeting 28th August, 1995 has appointed a Committee of Officials to review the reservation list and submit recommendations to the Advisory Committee from time to time.
- Explanations range from competition from imports, technology requirements, needs to comply with regulation, no benefit to small producers or availability of unreserved substitute products.
- **Biscuits**
  - Monopoly: Only 5–6 oligopolistic manufactures against the interests of consumers.
  - Market Segmentation
  - Competition promote technology upgradation
  - Interest of Wheat Growers
- **Flooring tiles**
  - High value item having considerable scope of export.
  - Need for avoiding wastage by making maximum use of granite for manufacturing of tiles

Details

Identification Strategy

- **Dereservation**: Post 1991, with lower import tariffs, growing concern for small scale firms had to compete with imported goods
  - **Staggered removal** of reserved products from the reservation list using manufacturing firms in **India** Industry Distribution
    - **Oil:** Sesame Oil (2008) vs. Mustard Oil (2015)
    - **Leather:** Leather Shoes(2001) vs. Leather Slipper (2003)
    - **Dental:** Tooth Paste(2002) vs. Tooth Powder(2008)

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- **Identification Strategy:** Short windows around removal of entry barrier
  - Time gap between regulation and start of manufacturing (no change in industry structure)
  - Identify Threat of Entry vs. Actual Entry
  - Period before the entrant undertakes any sunk costs
  - Martin, Nataraj, and Harrison (AER, 2016) positive impact on employment growth after the removal of entry barrier

Incumbents' Reaction

Business Standard

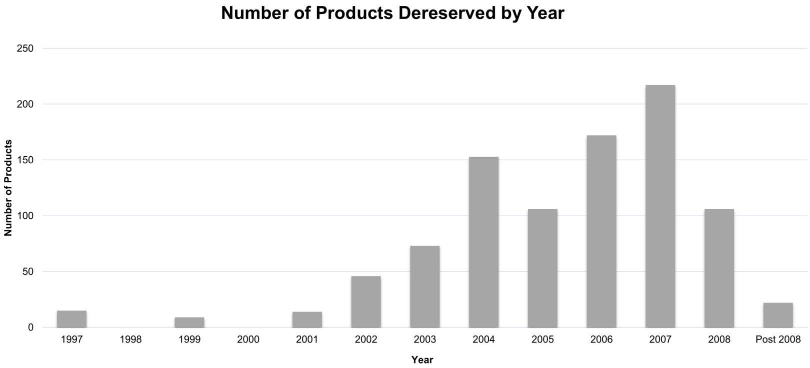
Ssis Feel Threatened By Mncs, Dereservation: Ficci Survey

BSCAL
December 29, 1997 Last Updated at 00:00 IST

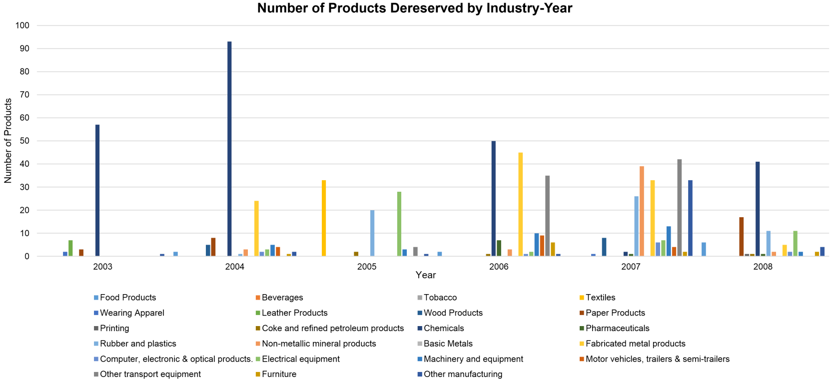
Small scale industries (SSIs) see the entry of multinational corporations (MNCs) and dereservation of the items reserved for them as a threat to their marketshare.

In a survey conducted by the Federation of Indian Chambers of Commerce & Industry (Ficci), majority of the respondents advocated gradual de-reservation of the sector as they do not enjoy the financial clout and infrastructure network of large business houses and MNCs.

Dereservation by Year



Dereservation by Industry-Year



Back

Data

- **Data Period:** 1991 to 2013
- **Data Source:** Centre for Monitoring the Indian Economy (CMIE) Prowess Database contains information primarily from the income statements and balance sheets of about more than 5000 firms in manufacturing sector
- **Product Data:** There are 2,710 products, defined as five-digit National Industrial Classification (NIC) codes linked to 137 four-digit NIC industries
- **Price Data**: For each product manufactured by the firm, the data set provides the value of sales, quantity, and units. I create price index using this information.
- **Hand Match:** Reserved/Dereserved Products with NIC5 digit product code

Examples

Product Data

	Firm 1 Sales Revenue	Firm 2 Sales Revenue	Treatment Group
 Brake Pad	70%	30%	1
 Seat Covers	30%	70%	0

Methodology

$$y_{it}=\alpha_0+\sum_{k=-2}^{-10}\beta_k Pre-Dereservation(k)_{it}+\sum_{k=0}^{10}\beta_k Dereservation(k)_{it}+\alpha_i+\alpha_t+Inc_i*\alpha_t+\varepsilon_{it}$$

- $\mathbf{y}_{it}$ is defined as the *accounts receivables scaled by sales* of firm i at time t
- **Treatment:** Elimination of small-scale reservation on the incumbent firm's first-observed **primary** reserved product in NIC-5 digit category. Firm's primary product is based on **maximum sales revenue**.
- **Control:** Pre-Dereservation observation of firms treated later and firms in same industry never produced reserved/dereserved product
- *Pre-Dereservation(k)$_{it}$* (*Dereservation(k)$_{it}$*) is a dummy variable that takes a value one if it is k years before (after) the incumbent firm's primary reserved product has been dereserved.
- An **incumbent** firm is defined as a firm that has **ever made** a reserved product before it was dereserved.
- Standard errors are corrected for heteroscedasticity and auto correlation, and clustered at NIC5-digit product level.

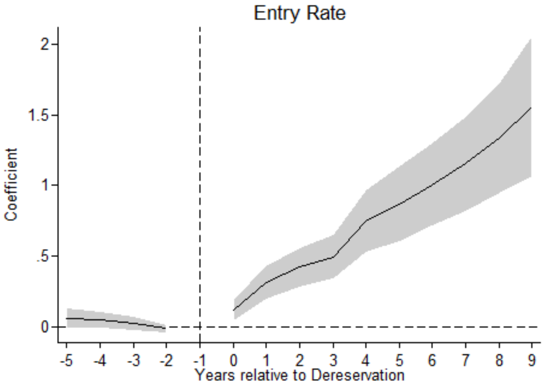
Effect of Firm characteristics on Timing of Dereservation

Cox Survival Regressions - Exclusion Restriction

	Time to Dereservation					
	(1)	(2)	(3)	(4)	(5)	(6)
Trade Credit Granted(Industry Avg.)	-1.240 (1.978)					3.683 (3.826)
Firm Size (Industry Avg.)		.251 (.312)				.387 (.286)
Profitability(Industry Avg.)			4.588 (4.875)			4.267 (3.685)
Cash Holdings(Industry Avg.)				.988 (2.296)		1.181 (3.319)
Sales Growth(Industry Avg.)					1.330 (1.142)	2.071 (1.307)
Industry Fixed Effects	4-digit	4-digit	4-digit	4-digit	4-digit	4-digit
Obs.	1065	1065	1065	1065	1065	1065
Log Pseudo likelihood	-241.64	-241.309	-241.303	-241.692	-240.477	-238.875
χ^2 statistic	16884.3	12924.58	1887.563	56474.67	1892.95	16223.03

Effect of Dereservation on Industry Competition

Difference-in-Differences, Strength of Instrument



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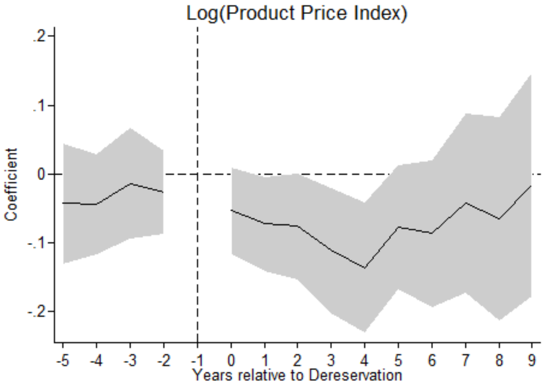

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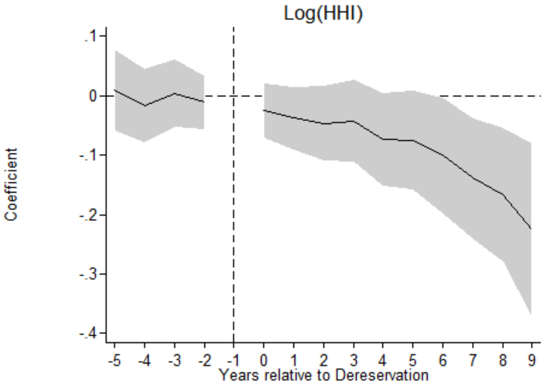
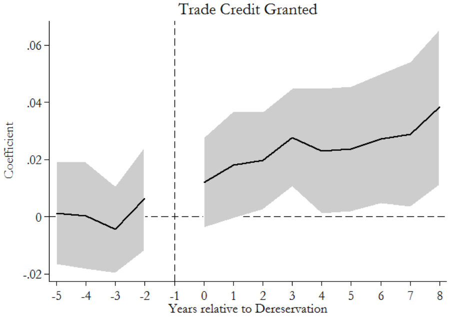


Base Line Results

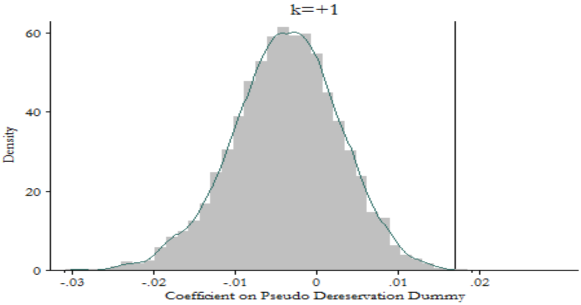


- Average incumbent firm extends 11% more trade credit (avg. 19%)
- Equivalent to increase in term by 10-12 days (avg. 70 days)
- Accounts receivables increase by about INR 47 mn. ($0.78 mn.) , avg. firm size ($40 mn.)

Placebo Test

Non-Parametric Distribution of Placebo Dereservation Estimates

Randomized assignment of dereservation years to similar products, keeping the distribution of the event years unchanged



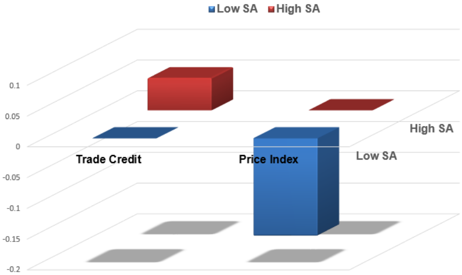
Price vs. Trade Credit Supply

- Price Index of dereserved products goes down for effected firms
- Most of the effect comes after 4-5 years, when entry starts
- Prices go down by almost 8.6%

Robustness Checks

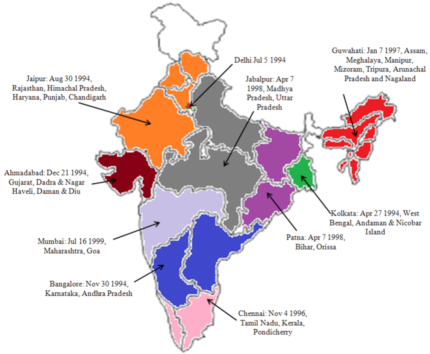
- Only Treatment Firms
- State-year Fixed Effects
- Excluding Export Firms
- Alternative Definition of Incumbent Firms
- Alternative Definition of Trade Credit
- Drop industries with Large Group Firms

Access to Finance: Size-Age Index



- *Cut based on previous year Size-Age Index*
- **High Size-Age Index**
  - Trade Credit, +9.3 pp. (avg. 19 %)
  - Price Index, No Change
- **Low Size-Age Index**
  - Trade Credit, No Change
  - Price Index, -16.0%

Access to Finance: Debt Recovery Tribunals(DRTs)

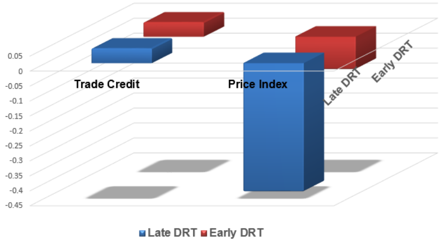


- *Quasi-natural experiment i.e. establishment of Debt Recovery Tribunals (DRTs) in India*

- Lower enforcement costs
  - Size of New Loans (+)
  - Debt Maturity (+)
  - Interest costs (-)

  - Visaria, (2009, 2012), Gopalan, Mukherjee, and Singh (2016)

Access to Finance: Debt Recovery Tribunals(DRTs)



- *Cut based on firms located in early vs. late DRT states*

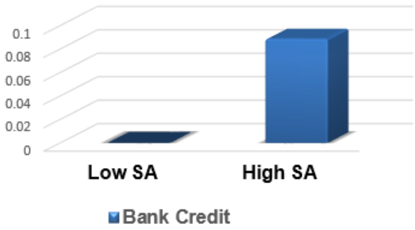
- **Early DRT**
  - Trade Credit, +5.0 pp. (avg. 19 %)
  - Price Index, -11.4 % in first year

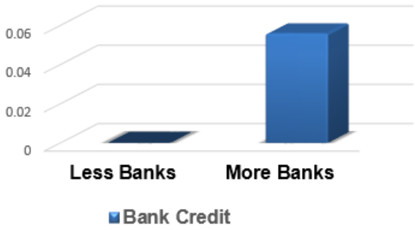
- **Late DRT**
  - Trade Credit, 5.1 % after three years
  - Price Index, -43.0% in first year

Access to Finance: Robustness Checks

- Number of banking relationships
- Relationship with large banks
- State-level access to banks
- Access to supplier's credit

Financing Trade Credit





Dereservation of Input

- Trade Credit Received/Price with dereservation firm's major input
- Trade Credit Received (+)
- Prices (-)
- Low Size-Age and late DRT firms receive more trade credit
- Consistent with financially constrained customers value trade credit instead of price cuts.

Conclusions

- Do firms extend more credit to their customers when they face entry threat from competitors ?
  - Yes !
- Use, *plausibly* exogenous, *staggered* removal of product level entry barriers over time for Indian manufacturing firms
  - Increase in competition which reduced market concentration by 10% in affected industries in the subsequent 5-6 years
  - Incumbent firm extends 8-10% more trade credit when threat of entry increases, especially those with deep pockets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"But Mom, if I apologize to my competitors, it takes all the fun out of capitalism!"

Thank You !